Q2A | Summarize the Data

> A. Summarize the data. How much data is present? What attributes/features are continuous valued? Which attributes are categorical?

This notebook answers the question in three parts:

1. **How much data is present?** Rows, columns, missing values, and duplicates.
2. **Which attributes are continuous?** Data types, distinct values, value ranges, and the date column.
3. **Which attributes are categorical?** The values each categorical attribute takes, and the one column that is neither.

IMPORTANT

The dataset is a single file with no `train` / `test` split. The split is made in a later notebook, so this summary describes the whole dataset.

Answer Summary

- **How much data is present?** 731 rows and 16 columns (the target `Rentals` and 15 others), with no missing values and no duplicate rows. Each row is one day, from 2011-01-01 to 2012-12-31.
- **Which attributes are continuous?** 8: `Temp`, `FeelsLike`, `Humidity`, and `Wind` (decimals between 0 and 1), `Casual`, `Registered`, and `Rentals` (whole-number counts with 606 to 696 distinct values), and `Date` (it measures time, so the gap between two dates is an amount of days).
- **Which attributes are categorical?** 7: `Season`, `Year`, `Month`, `IsHoliday`, `Weekday`, `IsWorkday`, and `Weather`, each with 2 to 12 distinct values. `ID` is neither: it is a label for the row.

Flags:

- The rows are in date order, one per day with no gaps, which matters for how the data is split.
- `Date` is not stored as a number, and `ID` carries the same information as `Date`.
- `Casual` and `Registered` add up to `Rentals`, which matters for whether they can be used as attributes.
- The categorical attributes are stored as integer codes, and `IsHoliday` and `Weather` each have a value with only 21 rows.

- The scales differ: 0 to 1 for the measurements, up to 12 for the codes, thousands for the counts.
- `Humidity` is the only decimal column with a minimum of 0, a value to check as a possible error.

Imports

In [1]:
```
import pandas as pd
from IPython.display import display

import Q2.consts as Q2_CONSTS

from utils import read_csv_to_df
```

Fetch Data

In [2]:
```
df = read_csv_to_df(path=Q2_CONSTS.DATA_PATH, parse_dates=["dteday"])
df = df.rename(columns=Q2_CONSTS.SHORT_NAMES)
```

1. How Much Data is Present?

In [3]:
```
summary = pd.DataFrame(
        {
                "row count": [ df.shape[0] ],
                "columns count": [ df.shape[1] ],
                "missing values": [ df.isnull().sum().sum() ],
                "rows w/ missing values": [ df.isnull().any(axis=1).sum()
],
                "% rows w/ missing values": [
                        round(df.isnull().any(axis=1).sum() / df.shape[0] *
100, 2),
                ]
        },
        index=["Full Dataset"],
        dtype=object,  # keeps the counts as whole numbers after the
transpose
)

summary.T
```

Out[3]:

	Full Dataset
row count	731
columns count	16
missing values	0
rows w/ missing values	0
% rows w/ missing values	0.0

- 16 columns: the target `Rentals` and 15 others. No value is missing, so there is nothing to impute.

Duplicate Rows

In [4]:
```
df.duplicated().sum()
```

Out[4]:
```
0
```

- No duplicate rows, so there is nothing to remove.

2. Which Attributes are Continuous?

An attribute is continuous if it measures an amount that can take many values within a range. The stored data type does not settle this, so the evidence is the number of distinct values and the range of each column.

In [5]:
```
info_df = pd.DataFrame({
        "dtype": df.dtypes,
        "unique values": df.nunique(),
        "most frequent value count": df.apply(lambda col:
col.value_counts().iloc[0]),
}).sort_values(by=["unique values", "most frequent value count"],
ascending=False)

info_df
```

Out[5]:

	dtype	unique values	most frequent value count
ID	int64	731	1
Date	datetime64[ns]	731	1
Rentals	int64	696	2
FeelsLike	float64	690	4
Registered	int64	679	3
Wind	float64	650	3
Casual	int64	606	4
Humidity	float64	595	4
Temp	float64	499	5
Month	int64	12	62
Weekday	int64	7	105
Season	int64	4	188
Weather	int64	3	463
IsHoliday	int64	2	710
IsWorkday	int64	2	500
Year	int64	2	366

- The distinct value counts split the 16 columns into two groups with nothing in between: 7 columns with 2 to 12 distinct values, and 9 columns with 499 to 731 where no value repeats more than 5 times.
- All 7 columns in the first group are `int64`, like the counts `Casual`, `Registered`, and `Rentals`, so the data type does not tell the two groups apart.
- `Date` is the only non-numeric column (`datetime64`).

Value Ranges

In [6]:

```
numeric_df = df.select_dtypes(include="number")

value_ranges = pd.DataFrame({
        # object dtype keeps whole-number columns without decimals
        "min": pd.Series({col: round(numeric_df[col].min(), 3) for col in
numeric_df}, dtype=object),
        "max": pd.Series({col: round(numeric_df[col].max(), 3) for col in
numeric_df}, dtype=object),
        "mean": numeric_df.mean().round(3),
        "std": numeric_df.std().round(3),
        "whole numbers only": numeric_df.apply(lambda col: (col.dropna() %
1 == 0).all()),
}, index=numeric_df.columns)
display(value_ranges)

pd.DataFrame(
        {
                "rows where Casual + Registered = Rentals": [ (df["Casual"]
+ df["Registered"] == df["Rentals"]).sum() ],
```

```
        },
        index=["Full Dataset"],
).T
```

	min	max	mean	std	whole numbers only
ID	1	731	366.000	211.166	True
Season	1	4	2.497	1.111	True
Year	0	1	0.501	0.500	True
Month	1	12	6.520	3.452	True
IsHoliday	0	1	0.029	0.167	True
Weekday	0	6	2.997	2.005	True
IsWorkday	0	1	0.684	0.465	True
Weather	1	3	1.395	0.545	True
Temp	0.059	0.862	0.495	0.183	False
FeelsLike	0.079	0.841	0.474	0.163	False
Humidity	0.0	0.972	0.628	0.142	False
Wind	0.022	0.507	0.190	0.077	False
Casual	2	3410	848.176	686.622	True
Registered	20	6946	3656.172	1560.256	True
Rentals	22	8714	4504.349	1937.211	True

Out[6]:

	Full Dataset
rows where Casual + Registered = Rentals	731

- The scales differ, which matters for SGD and regularization: the 4 decimal columns lie between 0 and 1, while the counts reach the thousands.
- `Casual` and `Registered` add up to `Rentals` in all 731 rows (the check table right above), as two parts of the target would. This matters for whether they can be used as attributes.
- `Humidity` is the only decimal column with a minimum of 0, a value to check as a possible error.

7 of the 15 numeric columns are continuous: the 4 decimal columns (measurements) and the 3 rental counts, which are whole numbers but are amounts with 606 to 696 distinct values.

Date

In [7]:
```
days_since_first = (df["Date"] - df["Date"].min()).dt.days

date_summary = pd.DataFrame(
        {
                "first date": [ df["Date"].min().date() ],
                "last date": [ df["Date"].max().date() ],
                "distinct dates": [ df["Date"].nunique() ],
```

```
                "days from first to last date": [ days_since_first.max() ],
                "rows 1 day after the previous row": [
(df["Date"].diff().dt.days == 1).sum() ],
                "rows where ID = days since first date + 1": [ (df["ID"] ==
days_since_first + 1).sum() ],
        },
        index=["Date"],
)

date_summary.T
```

Out[7]:

	Date
first date	2011-01-01
last date	2012-12-31
distinct dates	731
days from first to last date	730
rows 1 day after the previous row	730
rows where ID = days since first date + 1	731

- The rows are in date order, one per day with no gaps (all 730 rows after the first are 1 day after the row before), which matters for how the data is split.
- `Date` measures time, so it is continuous: counted as days since the first date it is an amount from 0 to 730, and the gap between two dates means something.
- `ID` is a label for the row, so it is neither continuous nor categorical. It equals that day count plus 1 in all 731 rows only because the file is in date order with no gaps, so it carries the same information as `Date`.

8 attributes are continuous: the 4 measurements (`Temp`, `FeelsLike`, `Humidity`, and `Wind`), the 3 rental counts (`Casual`, `Registered`, and the target `Rentals`), and `Date`.

3. Which Attributes are Categorical?

An attribute is categorical if its values name a group that a row belongs to: the value is a label, not an amount. The candidates are the 7 columns with 12 or fewer distinct values from section 2.

In [8]:
```
categorical_cols = ["Season", "Year", "Month", "IsHoliday", "Weekday",
"IsWorkday", "Weather"]

category_levels = pd.DataFrame({
        "distinct values": df[categorical_cols].nunique(),
        "values": [", ".join(map(str, sorted(df[col].unique()))) for col in
categorical_cols],
        "fewest rows per value": [df[col].value_counts().min() for col in
categorical_cols],
        "most rows per value": [df[col].value_counts().max() for col in
```

```
categorical_cols],
})

category_levels
```

Out[8]:

	distinct values	values	fewest rows per value	most rows per value
Season	4	1, 2, 3, 4	178	188
Year	2	0, 1	365	366
Month	12	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12	57	62
IsHoliday	2	0, 1	21	710
Weekday	7	0, 1, 2, 3, 4, 5, 6	104	105
IsWorkday	2	0, 1	231	500
Weather	3	1, 2, 3	21	463

- All 7 take a few whole-number values, and their names describe groups of days, so the numbers are labels, not amounts. They are stored as integer codes, which matters for how they are encoded.
- `IsHoliday` and `Weather` each have a value with only 21 of the 731 rows, which matters for any split of the rows.

7 attributes are categorical: `Season`, `Year`, `Month`, `IsHoliday`, `Weekday`, `IsWorkday`, and `Weather`.

1 column is neither continuous nor categorical: `ID`, the row number (see the Date section).